Department of Computer Systems Engineering
Faculty of Computing
Sri Lanka Institute of Information Technology

IE4060: Robotics and Intelligent Systems

Year 04 Semester 01/02

2026

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| Assignment Title | **Autonomous Navigation of TurtleBot3 Burger in Gazebo (ROS2 Jazzy)** |
| Learning outcomes covered | **LO 02:** Demonstrate the knowledge and understanding of the theories, techniques, mathematical models and methods involved in mobile robot motion.<br>**LO 03:** Plan and program mobile robot actions by applying a range of artificial intelligence techniques and control paradigms.<br>**LO 04:** Describe and analyse the general purpose component and middleware technologies that support many of the fundamental architectural patterns used in robots and other similar complex systems.<br>**LO 05:** Demonstrate the knowledge and understanding of Human-Robot Interaction models used for robot design.<br>**LO 06:** Work in project teams to develop software to meet changing requirements for a complex application |
| Assignment Mode | Project |
| Maximum Marks | 100 |
| Contribution to the Final Grade | 30% |
| Date published | |
| Deadline for submissions | 18th October 2026 |
| Mode of Submission | Project Submission |

**Autonomous Navigation of TurtleBot3 Burger in Gazebo (ROS2 Jazzy)**

**Introduction**

In this project, you will design and implement an autonomous navigation system for the TurtleBot3 Burger robot in a maze-like environment using Gazebo and ROS 2 Jazzy. The goal is to allow the robot to reach any selected point in the maze with minimal travel time while avoiding obstacles.

This project will challenge you to integrate concepts from robotics, intelligent systems, navigation, and machine learning.

**Project Objectives**

- Create a custom maze environment in Gazebo.
- Use TurtleBot3 Burger with LiDAR-based SLAM or equivalent mapping.
- Implement autonomous navigation.
- Enable user input (goal selection by clicking a location on the map).
- Ensure the robot reaches the goal efficiently and safely.

**Tasks**

Environment Creation

- Build a maze world in Gazebo with as given in *Figure 1*.
- Refer: https://www.youtube.com/watch?v=J_vasKVhUQM&t=174s

Mapping & Localization

- Use SLAM (Simultaneous Localization and Mapping) to map the environment.

Navigation & Planning

- Implement autonomous navigation using Nav2 stack (Dijkstra/A*) or a custom path planning algorithm.
- Implement a machine learning approach (Deep Q-learning).

Goal Execution

- The user should be able to click a destination point.
- The robot must autonomously move to the goal, avoiding collisions.

Performance Optimization

- Minimize travel time.
- Ensure smooth and efficient navigation.

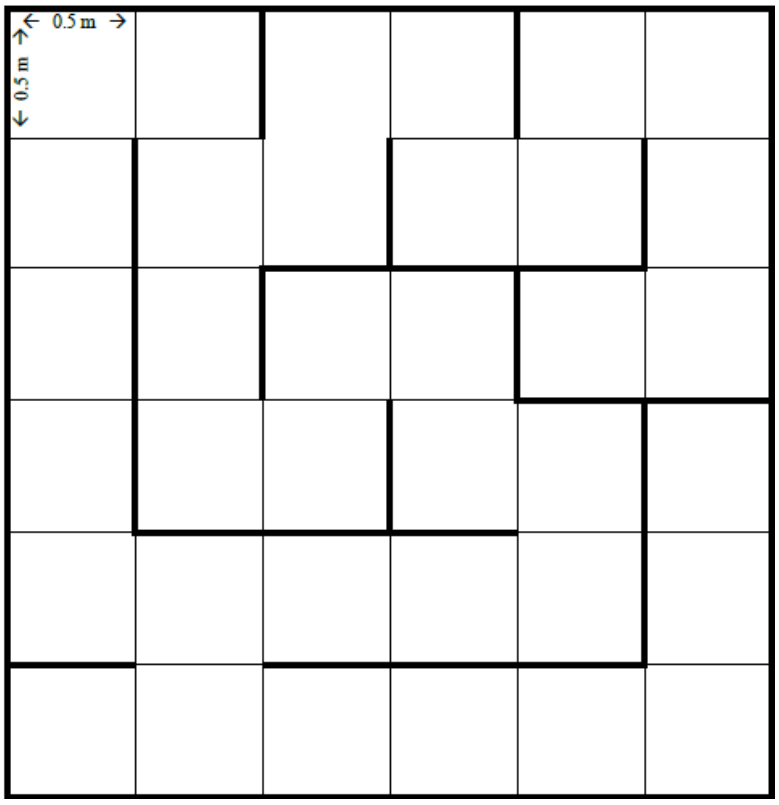



*Figure 1: Robot Maze*

**Deliverables**

- Simulation World File
  - Gazebo maze world (.world file).
- Implementation Files
  - ROS 2 packages, launch files, and configuration (submitted as GitHub repo).
- Final Report (5–7 pages)
  - Maze design explanation.
  - Navigation method used.
  - Results (travel time, path efficiency in different methods used, challenges faced).
  - Possible improvements.
- Demonstration (At the VIVA session)
  - Show the robot navigating to at least three different clicked goal points in the maze.

**Evaluation Criteria**

Maze creation in Gazebo - 10%

SLAM / Mapping implementation - 20%

Autonomous navigation working - 35%

Efficiency (time, path length, obstacle avoidance) - 10%

Report (clarity, results, explanation) - 10%

Demonstration and VIVA - 15%

**Submission Instructions**

- Submit your maze world file, ROS 2 packages to the GitHub repo (Share the link in cover page of the report) and submit the report to CourseWeb by the due date.
- Report Submission Deadline: 18th October 2026, 11.59 PM
- VIVA Data: 18th October (Lecture Time)

** End of Assignment **